

Atharvaa Rane

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SUMMARY

Software Engineer with around 4 years of experience building AI-enabled and data-driven Python applications, with a strong focus on scalability, performance, and production-ready machine learning systems. Experienced in developing backend services and ML pipelines using Python, Django, and Flask, and integrating machine learning and NLP models into enterprise applications. Skilled in data preprocessing, feature engineering, model validation, and ML inference services supporting real-time analytics and decision automation. Proficient in designing and managing relational and NoSQL databases including MySQL, PostgreSQL, SQL Server, MongoDB, and Firebase. Hands-on experience in building RESTful APIs to expose AI/ML capabilities and deploying containerized applications on AWS, Azure, and Databricks.

SKILLS

Methodology: SDLC, Agile, Waterfall

Programming Languages: Python, SQL, PL/SQL, Java, C#, ASP.NET, C, C++

IDEs: Visual Studio Code, Jupyter Notebook, PyCharm

Machine Learning & AI: Supervised & Unsupervised Learning, Regression, Classification, Clustering, Time-Series Forecasting, Feature Engineering, Model Evaluation, Hyperparameter Tuning, Generative AI, Reinforcement Learning, Fine-Tuning, RLHF, A/B Testing & Experimentation, Responsible AI, Model Quantization & Optimization

AI / LLM Models: GPT-OSS, Open-Source GPT Models, Transformer-Based Models, LLM Evaluation (Evals), Prompt Engineering, Few-Shot & Zero-Shot Learning, Multi-Modal AI, RAG (Retrieval-Augmented Generation), Embeddings & Vector Search

LLM Frameworks & Tooling: LangChain, LangSmith, Hugging Face Transformers, OpenAI API, Anthropic API, ONNX, Triton Inference Server

ML Frameworks & Libraries: Scikit-learn, TensorFlow, PyTorch, JAX, NumPy, Pandas, XGBoost, LightGBM, Web2py, CherryPy, Requests, OpenCV

MLOps & Experiment Tracking: MLflow, Weights & Biases (W&B), Kubeflow, Apache Airflow, CI/CD for ML Pipelines (GitHub Actions, Jenkins), Experiment Tracking & Model Versioning

Natural Language Processing (NLP): Text Classification, Sentiment Analysis, Topic Modeling, Named Entity Recognition

Frameworks: Django, Flask, FastAPI, Node.js, Express.js, React.js, Angular, .NET Framework

Cloud & Infrastructure: AWS (SageMaker, EC2, S3), Azure (Azure ML), GCP (Vertex AI), Databricks, Docker, Kubernetes, Helm, Terraform

Data & Warehousing: MySQL, SQL Server, MongoDB, Firebase, PostgreSQL, Snowflake, BigQuery, Redshift, Redis, Apache Spark, Kafka, dbt

Vector Databases: Pinecone, Weaviate, ChromaDB

Web Technologies: HTML, CSS, JavaScript, AJAX, XML, JSON, jQuery, Bootstrap, RESTful API, Redux

Version Control / Other Tools: GitHub, Git, GitLab, Jira, Looker / BI Tools

Operating Systems: Windows, Linux, Mac

EXPERIENCE

COSMOS Research Center at UALR | Aug 2025 - Present

Software Engineer (AI)

- Designed, deployed, and owned production-grade AI applications using Python, integrating machine learning, NLP, and Generative AI (LLMs) to deliver scalable analytics and automation solutions.
- Productionized end-to-end ML pipelines (regression, classification, clustering, time-series forecasting) using Scikit-learn, XGBoost, and LightGBM, improving model accuracy by 20% and supporting repeatable experimentation.
- Implemented LLM-powered services using open-source GPT models to enable text classification, sentiment analysis, and automated insight generation at scale.
- Built and maintained RESTful AI inference services to support real-time predictions and seamless integration with downstream research and data platforms.
- Containerized and operationalized ML and LLM workflows using Docker, improving environment reproducibility and reducing deployment cycle time by 30%.

HCL Tech, USA | Nov 2024 - Jul 2025

Software Engineer

- Designed and deployed LLM-powered applications using RAG architectures with LangChain, OpenAI, and vector databases (Pinecone, ChromaDB), improving response relevance by 40% and enabling intelligent automation across enterprise workflows.
- Built scalable backend services using FastAPI, Flask, and microservices architecture to serve AI/ML models, achieving sub-100ms latency and supporting high-concurrency production environments.
- Developed and fine-tuned transformer-based models using PyTorch and Hugging Face, improving NLP task performance (classification, NER) by 25% across multiple datasets.
- Implemented advanced prompt engineering and evaluation pipelines (few-shot, zero-shot, A/B testing), improving LLM output quality and reducing hallucinations by 30%.
- Built end-to-end ML pipelines using Airflow, MLflow, and Kubeflow, enabling automated training, experiment tracking, and model versioning, reducing deployment time by 50%.
- Optimized model inference using ONNX and Triton Inference Server with quantization techniques, reducing latency by 45% and improving throughput in production systems.
- Integrated cloud-based AI solutions using AWS and Azure ML, improving scalability and reducing infrastructure costs by 20%.

- Developed full-stack applications using React.js and Node.js to deliver AI-powered user interfaces, improving user engagement and enabling seamless interaction with ML systems.
- Implemented monitoring and observability using MLflow and logging frameworks, improving model performance tracking and reducing production issues by 30%.

Mindtree, India | Jan 2021 - Jul 2023

Software Engineer

- Developed and deployed machine learning models (classification, regression, clustering) using Scikit-learn and TensorFlow, improving prediction accuracy by 20% across multiple business use cases.
- Built and integrated NLP solutions using Hugging Face Transformers for tasks such as text classification and entity recognition, improving processing efficiency by 25% in production workflows.
- Implemented early-stage LLM applications using prompt engineering and RAG concepts, enabling intelligent document search and improving information retrieval accuracy by 30%.
- Designed and developed RESTful APIs using Flask and FastAPI to serve ML models, improving system scalability and reducing response time by 20% for downstream applications.
- Built data processing pipelines using Python, Pandas, and SQL, reducing data preparation time by 30% and improving data quality for analytics and modeling.
- Worked with Apache Spark and Kafka to handle large-scale data ingestion and processing, improving data throughput and enabling near real-time analytics capabilities.
- Executed model tracking and experiment management using MLflow, improving reproducibility and reducing model deployment issues by 25%.
- Deployed applications using Docker and supported CI/CD pipelines with Jenkins and Git, improving deployment efficiency by 35% and ensuring consistent releases.
- Developed basic front-end components using React.js and JavaScript to support AI-driven features, improving usability and interaction with ML-powered systems.

EDUCATION

Master of Science in Computer Science | Syracuse University, NY

May 2025

PROJECT

Task Tracker Web App — React, Node.js, PostgreSQL, REST APIs, Docker, GitHub Actions, JWT Authentication

- Launched a full-stack app for 500+ users with REST APIs and PostgreSQL; Docker/GitHub Actions cut deployment time 40% Smart Personalization API (Backend Infrastructure Project) — Node.js, AWS Lambda, GraphQL, CloudWatch, CI/CD
- Designed and deployed ML-driven personalization APIs using Node.js, AWS Lambda, and GraphQL, supporting low-latency inference in production

Syracuse City Infrastructure Forecasting — Python, Geospatial Analytics, Time-Series Forecasting, ARIMA, LSTM

- Built ARIMA and GeoPandas models integrating time-series and geospatial data, reducing service delays 15% and allocation accuracy 25%
- Financial Document & News Text Analysis Project - Python, scikit-learn, NLTK, pandas, NumPy, Matplotlib/Seaborn for visualization
- Built a Random Forest sentiment classifier (92.18% accuracy) for 500K+ financial articles; visualized insights for investment analysis

CERTIFICATIONS

- CompTIA Security+ | AWS Cloud Quest: Cloud Practitioner | Snowflake: Hands on Data Warehousing | Passed July 2025

PUBLICATIONS

- AI-driven Chatbot and its Evolution – Published in IEEE; focused on NLP-based conversational agent design & real-time deployment pipelines
- Real Time Sign Language Detection – Published in GIS Science Journal; developed a CNN-based classifier for ASL detection with 90%+ accuracy